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Case study: performance review of a 10 kWp hybrid plant in Hat Yai

18 October 2026 | Prepared by Solar ESS from remote monitoring data. Customer details removed.

Summary

The plant produces about two-thirds (63%) of what a 10 kWp array should in Hat Yai, most likely because only one of its two PV strings is working. The battery side works well: since it was added on 18 May 2026, the shop has covered 98% of its own electricity and buys almost nothing from the grid.

Two items need attention first:

1. **Only one PV input is producing.** All solar current enters on input 1; the second MPPT reads 0 V. Fixing this could add about 12-15 kWh a day.

2. **Export was blocked for 24 days (28 Mar - 20 Apr 2026).** The inverter held production down to the shop's own use, losing about 500 kWh.

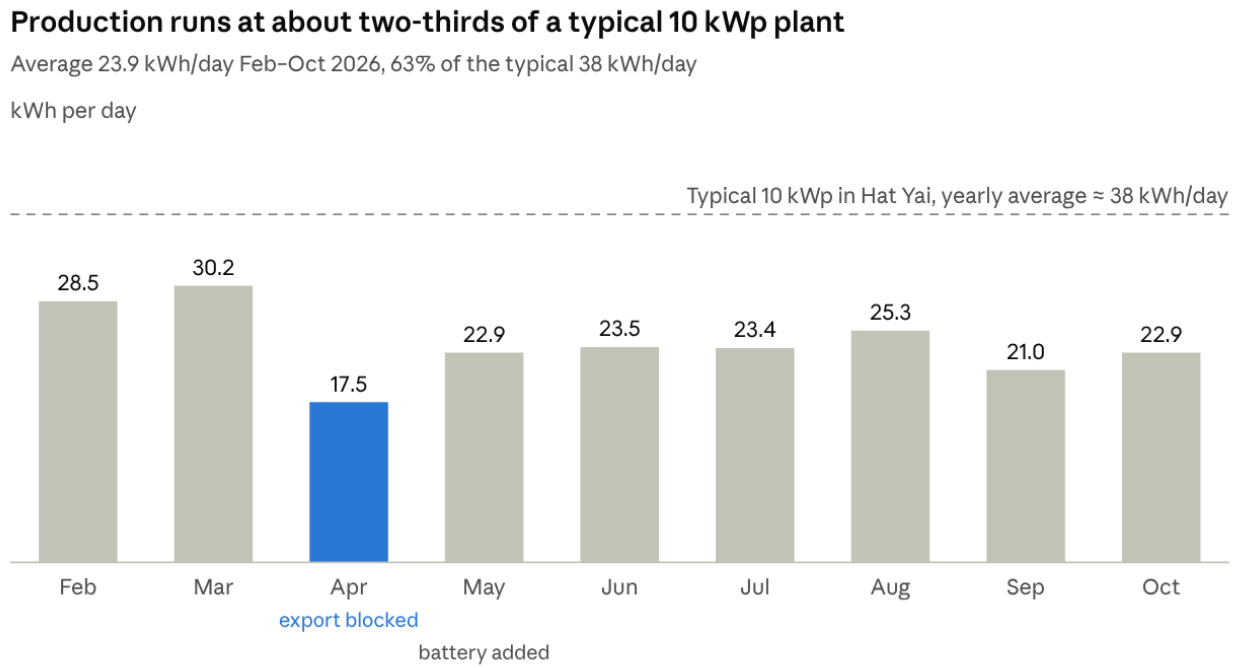
Plant at a glance

Review of all monitoring data from commissioning on 29 Jan 2026 to 17 Oct 2026 (262 days), from SolisCloud.

Item	Value
Location	Hat Yai, Songkhla
PV capacity (as registered)	10 kWp, tilt 30°
Inverter	Solis S6-EH3P10K hybrid, 10 kW, three-phase
Battery	One 48 V battery, added 18 May 2026
Total production	6,230 kWh (2.4 kWh per kWp per day)
Best day	36 kWh (25 Mar 2026)
Total consumption	6,183 kWh (about 22-23 kWh/day)
Grid import / export	1,402 kWh / 1,755 kWh
Self-sufficiency since battery	98% (import 0.5 kWh/day, was 12.4)
Active alarms	None

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Monthly performance



Source: SolisCloud daily totals, Feb-Oct 2026 (Oct = 1-17 Oct). The typical value is an approximate yearly average.

Production stays well below a typical 10 kWp plant in every month, dry season included. April is lowest because export was blocked for most of the month.

Findings

1. Only one PV input is producing

All solar current enters the inverter on PV input 1. MPPT2 reads 0 V and 0 A, and inputs 2-4 carry no current. At midday on sunny October days, output levels off at 4.6-5.3 kW, with 14-16 A at about 360 V. That is the current of one string; two strings in parallel would give about 26 A.

The best day so far, 36 kWh in the dry season, is about 3.6 kWh per kWp. A healthy 10 kWp array in Hat Yai should reach 45-50 kWh on a clear day. Either half the panels are not connected, or about 6 kWp is installed and the plant was registered as 10 kWp.

2. Export was blocked for 24 days

From 28 Mar to 20 Apr 2026 the plant exported nothing, and each day's production equalled the shop's daytime use. Production fell to 11 kWh/day, against 30-34 kWh/day in the two weeks before and after. That is about 500 kWh lost. The inverter's operation log in SolisCloud is empty, so the setting was changed on the inverter itself or from the installer's app.

3. Selling back to the grid

The plant has exported 1,755 kWh since January. Without a PEA sell-back contract, this energy earns nothing. Export will grow once item 1 is fixed, because the battery already covers the night.

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4. Battery energy counters do not add up

Since May the battery shows 1,832 kWh discharged but only 1,526 kWh charged. A battery cannot give out more than it takes in, so its readings are wrong. SolisCloud lists the battery maker as "OTHER". If the inverter misreads the battery's charge level, the battery can be over-discharged or never fully charged.

5. Steady overnight load

The shop draws about 0.7 kW all night, every night: about 8-9 kWh, close to 40% of daily use. Every 0.3 kW cut saves about 2.6 kWh of battery each night.

6. Inverter runs hot

The inverter's internal temperature was 58 °C at only 2 kW output (18 Oct 2026, 11:37). Inverters reduce output when they get too hot.

7. Earnings figures in the app

SolisCloud values every kWh produced at a flat 4.7 THB and grid power at 0 THB. The earnings shown in the app are not meaningful until import and export prices are set.

Recommended actions

Priority	Action	Expected benefit
1	Site visit: count the panels, measure each string's open-circuit voltage, check the DC isolator, fuses and MC4 connectors; connect the second string to MPPT2	Up to about 12-15 kWh/day more, if 10 kWp is installed
2	Ask the installer why export was blocked from 28 Mar to 20 Apr, and lock the export setting	Prevents another loss like the 500 kWh in April
3	Apply for (or confirm) a PEA sell-back contract	Paid export instead of free export
4	Check the battery brand setting and the battery-to-inverter communication; recalibrate the charge level	Correct charge readings and longer battery life
5	List what runs overnight and switch off what isn't needed	About 2.6 kWh of battery saved per night for each 0.3 kW cut
6	Check the inverter's ventilation and shading from direct sun	Avoids output cuts on hot days
7	Set the import and export tariffs in SolisCloud	Correct earnings figures in the app

Notes on the data

• Source: SolisCloud daily totals since 29 Jan 2026, and 5-minute inverter data for 1 and 17 Oct 2026 (older 5-minute data is no longer stored).

• The expected 45-50 kWh on a clear day for 10 kWp is a typical figure for southern Thailand, not a site-specific simulation.

• Most of the lower production from May to October is due to the monsoon. Findings 1 and 2 are equipment and settings problems, not weather.

• Prepared by Solar ESS from remote monitoring only; findings 1, 4 and 6 should be confirmed on site.